

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Werner's Theory, IUPAC Nomenclature & Isomerism	19 PYQs	Coordination Principles
High	Valence Bond Theory (VBT) & Crystal Field Theory (CFT)	14 PYQs	Bonding in Complexes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Werner's Theory, IUPAC Nomenclature & Isomerism	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 + 2\text{AgNO}_3 \rightarrow 2\text{AgCl} \downarrow + [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+} + 2\text{Cl}^-$ (Only outer sphere ions precipitate)
Valence Bond Theory (VBT) & Crystal Field Theory (CFT)	$\Delta_t = \frac{4}{9}\Delta_o$ Strong Field: $\Delta_o > P \implies \text{Low Spin}$ Weak Field: $\Delta_o < P \implies \text{High Spin}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Werner's Theory, IUPAC Nomenclature & Isomerism (Very High Yield — 19 PYQs)

Core Principle: Werner's theory: Primary valency is ionizable (equals oxidation state, satisfied by negative ions); Secondary valency is non-ionizable (equals coordination number, satisfied by neutral molecules/anions, directional in space). Isomerism: Linkage (ambidentate ligands like $\text{NO}_2^- / \text{ONO}^-$), Ionization (exchange between inside/outside coordination sphere), Geometrical (cis/trans in square planar $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$), Optical (seen in octahedral complexes with bidentate ligands like $[\text{Co}(\text{en})_3]^{3+}$).

■ **Caution / JEE Pitfall:** Square planar complexes of type MA_4 , MA_3B , MA_2B_2 NEVER exhibit optical isomerism because they always possess a plane of symmetry!

■ **Speed Trick:** The number of moles of AgCl precipitate formed with excess AgNO_3 indicates directly the number of ionizable chloride ions outside the square brackets.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "How many ions per molecule are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "A ligand which has two different donor atoms and either of the two ligates with the central metal atom/ion in the complex is called _____." [Ans: D]
- [2025 Shift] Q#4 (MCQ): "In the complex ion $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, the co-ordination number of Fe is..." [Ans: N/A]

3.2 Valence Bond Theory (VBT) & Crystal Field Theory (CFT) (High Yield — 14 PYQs)

Core Principle: CFT: In an octahedral field, d-orbitals split into lower t_{2g} (d_{xy} , d_{yz} , d_{zx}) and higher e_g ($d_{x^2-y^2}$, d_{z^2}) by Δ_o . Tetrahedral splitting $\Delta_t = (4/9)\Delta_o$ (always weak field, no pairing). Spectrochemical series: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$. Strong field ligands (CN^- , CO) cause pairing ($\Delta_o > P$ implies low spin, inner orbital d^2sp^3). Weak field ligands (H_2O , F^-) do not pair ($\Delta_o < P$ implies high spin, outer orbital sp^3d^2).

■ **Caution / JEE Pitfall:** Tetrahedral complexes NEVER form low-spin complexes because crystal field splitting $\Delta_t = (4/9)\Delta_o$ is never large enough to overcome pairing energy.

■ **Speed Trick:** $[Ni(CO)_4]$: CO is a strong field ligand, pairs all electrons into $3d^{10}$ implies sp^3 tetrahedral and DIAMAGNETIC. $[Ni(CN)_4]^{2-}$ is dsp^2 square planar and DIAMAGNETIC.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Match List-I with List-II List-I (Complex) List-II (Geometry) (a) $[Co(NH_3)_6]^{3+}$ (i) Trigonal bipyramidal (b) $[Ni(CN)_4]^{2-}$ (ii) Tetrahedral (c) $[Fe(CO)_5]$ (iii) Square planar (d) $[NiCl_4]^{2-}$ (iv) Octahedral Choose the correct answer from the options given below." [Ans: N/A]
- **[2026 Shift] Q#2 (MCQ):** "Given below are two statements: Statement-I: The M-C σ bond is formed by the donation of lone pair of electrons on the carbon atom." Statement-II: The M-C π back bond is formed by the donation of lone pair of electrons from the carbon atom to the vacant d-orbitals of the metal." [Ans: N/A]
- **[2026 Shift] Q#4 (MCQ):** "Which of the following is the most stable complex ion?" [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II List-I (Complex) List-II (Geometry) (a) $[Co(NH_3)_6]^{3+}$ (i) Trigonal bipyramidal (b) $[NiCl_4]^{2-}$ (ii) Octahedral (c) $[Ni(CN)_4]^{2-}$ (iii) Tetrahedral (d) $[Fe(CO)_5]$ (iv) Square planar Choose the correct answer from the options given below.

- (A) a-ii, b-iii, c-iv, d-i
- (B) a-ii, b-i, c-iii, d-iv
- (C) a-iii, b-ii, c-iv, d-i
- (D) a-i, b-iii, c-iv, d-ii

Correct Answer: None

Solution Analysis: For each complex, we identify the coordination number and the usual geometry. $[Co(NH_3)_6]^{3+}$ has 6 ligands around cobalt. So its coordination number is 6, and complexes with coordination number 6 generally have octahedral geometry. (a) $\rightarrow$ (ii) $[NiCl_4]^{2-}$ has 4 chloride ligands around nickel. Here nickel is in +2 oxidation state, so it is a d^8 ion. Since Cl^- is a weak field ligand, the complex remains tetrahedral. (b) $\rightarrow$ (iii) $[Ni(CN)_4]^{2-}$ also has nickel in +2 oxidation state, so it is again d^8 . B

Benchmark Problem 2: KCET 2026 (Q#3)

How many ions per molecule are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?

- (A) 6
- (B) 4
- (C) 3
- (D) 2

Correct Answer: None

Solution Analysis: In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$, the two chloride ions are outside the coordination sphere. So in water, it dissociates as: $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2 \rightarrow [\text{Co}(\text{NH}_3)_6]^{2+} + 2\text{Cl}^-$. The complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$ remains as one ion, and the two chloride ions are separate. So total ions produced: $1+2=3$. Hence, the correct option is: Option C3